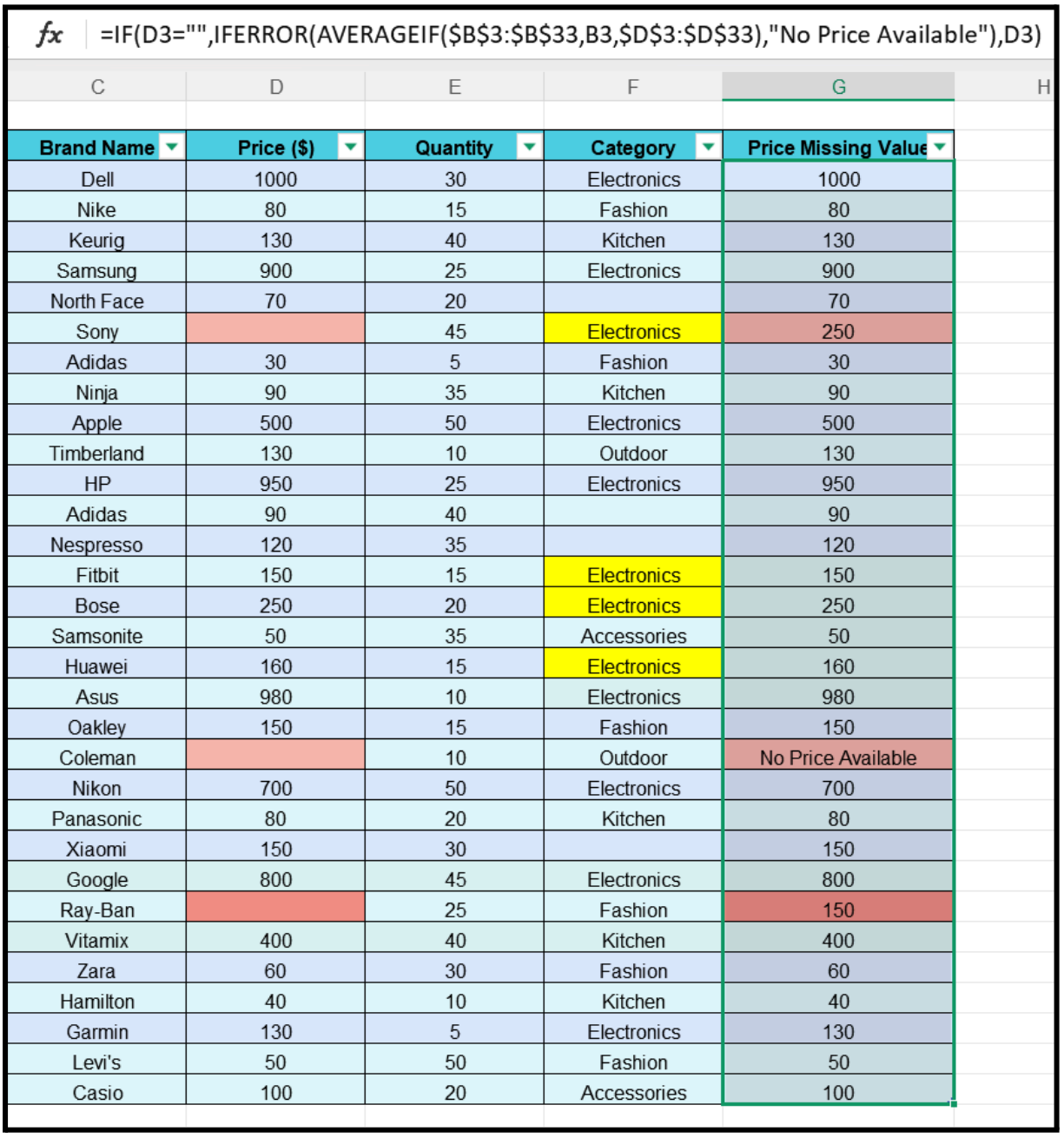
**Excel Assignment 2- Data Exploration**

## *1. Handling Missing Values*: *(Price Column)*



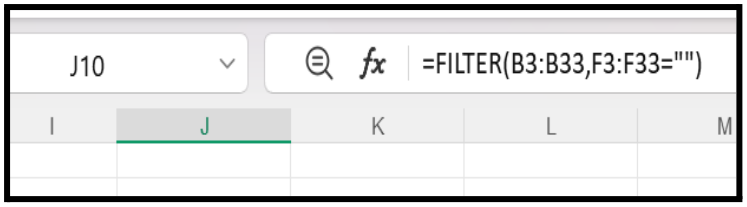
=IF(D3="",IFERROR(AVERAGEIF($B$3:$B$33,B3,$D$3:$D$33),"No Price Available"),D3)

| Brand Name | Price ($) | Quantity | Category | Price Missing Value |
|---|---|---|---|---|
| Dell | 1000 | 30 | Electronics | 1000 |
| Nike | 80 | 15 | Fashion | 80 |
| Keurig | 130 | 40 | Kitchen | 130 |
| Samsung | 900 | 25 | Electronics | 900 |
| North Face | 70 | 20 | | 70 |
| Sony | | 45 | Electronics | 250 |
| Adidas | 30 | 5 | Fashion | 30 |
| Ninja | 90 | 35 | Kitchen | 90 |
| Apple | 500 | 50 | Electronics | 500 |
| Timberland | 130 | 10 | Outdoor | 130 |
| HP | 950 | 25 | Electronics | 950 |
| Adidas | 90 | 40 | | 90 |
| Nespresso | 120 | 35 | | 120 |
| Fitbit | 150 | 15 | Electronics | 150 |
| Bose | 250 | 20 | Electronics | 250 |
| Samsonite | 50 | 35 | Accessories | 50 |
| Huawei | 160 | 15 | Electronics | 160 |
| Asus | 980 | 10 | Electronics | 980 |
| Oakley | 150 | 15 | Fashion | 150 |
| Coleman | | 10 | Outdoor | No Price Available |
| Nikon | 700 | 50 | Electronics | 700 |
| Panasonic | 80 | 20 | Kitchen | 80 |
| Xiaomi | 150 | 30 | | 150 |
| Google | 800 | 45 | Electronics | 800 |
| Ray-Ban | | 25 | Fashion | 150 |
| Vitamix | 400 | 40 | Kitchen | 400 |
| Zara | 60 | 30 | Fashion | 60 |
| Hamilton | 40 | 10 | Kitchen | 40 |
| Garmin | 130 | 5 | Electronics | 130 |
| Levi's | 50 | 50 | Fashion | 50 |
| Casio | 100 | 20 | Accessories | 100 |

The headphones and Sunglasses have two existing prices so it calculates an Average. But the Camping Tent has no existing prices so it cannot calculate an average (No price Values)

## 1. *Handling Missing Values: (Categories Column)*

Used IF, VLOOKUP, and IFERROR with a reference table to automatically identify and fill missing categories. Used FILTER it helps to find out which cells are empty.




H3

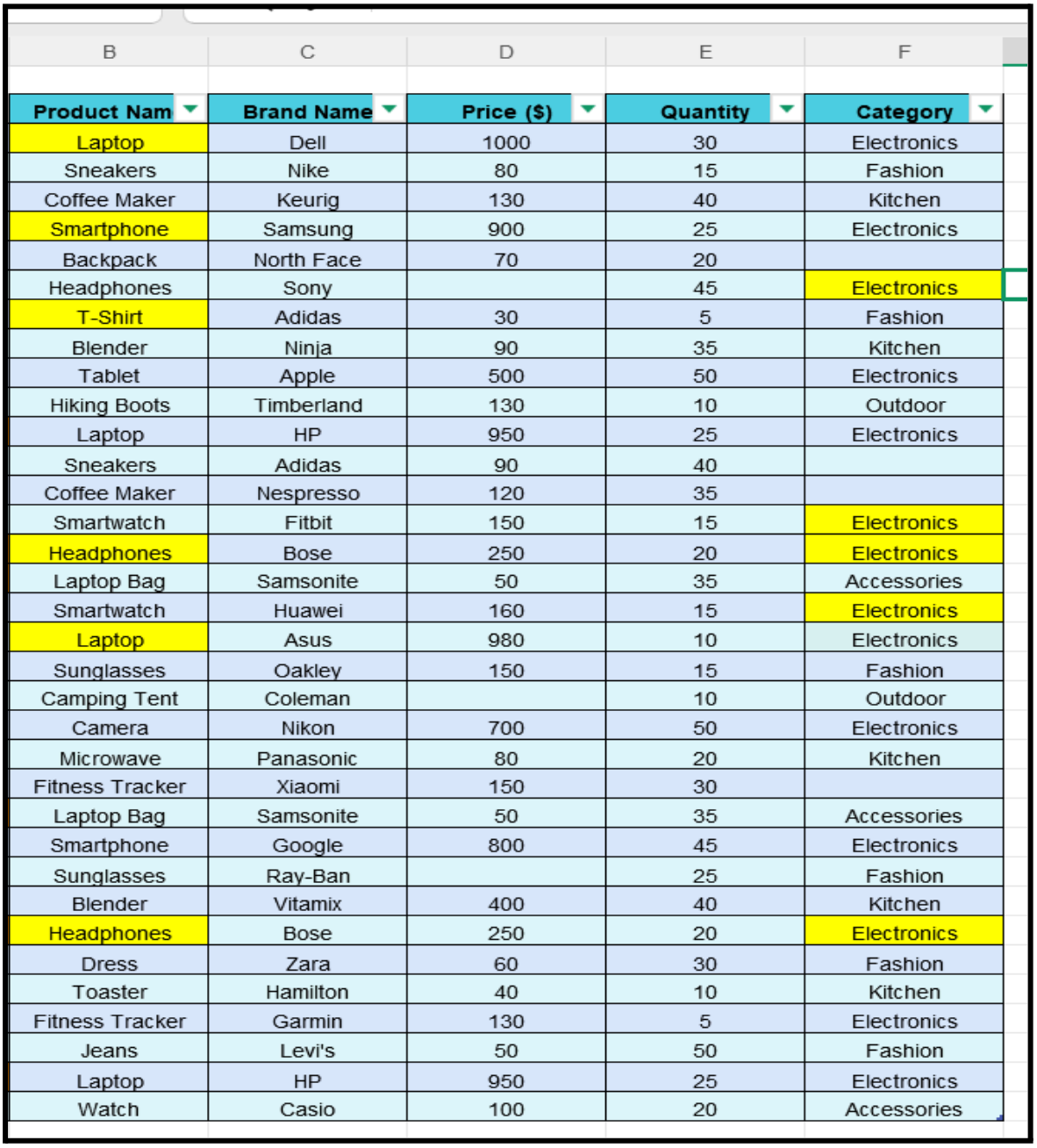
=IF(F3="",IFERROR(VLOOKUP(B3,$J$10:$K$13,2,FALSE),"Unknown"),F3)

| Category | Price Missing Value | Category Missing Value |
|---|---|---|
| Electronics | 1000 | Electronics |
| Fashion | 80 | Fashion |
| Kitchen | 130 | Kitchen |
| Electronics | 900 | Electronics |
| | 70 | Accesories |
| Electronics | 250 | Electronics |
| Fashion | 30 | Fashion |
| Kitchen | 90 | Kitchen |
| Electronics | 500 | Electronics |
| Outdoor | 130 | Outdoor |
| Electronics | 950 | Electronics |
| | 90 | Fashion |
| | 120 | Kitchen |
| Electronics | 150 | Electronics |
| Electronics | 250 | Electronics |
| Accessories | 50 | Accessories |
| Electronics | 160 | Electronics |
| Electronics | 980 | Electronics |
| Fashion | 150 | Fashion |
| Outdoor | No Price Available | Outdoor |
| Electronics | 700 | Electronics |
| Kitchen | 80 | Kitchen |
| | 150 | Electronics |
| Electronics | 800 | Electronics |
| Fashion | 150 | Fashion |
| Kitchen | 400 | Kitchen |
| Fashion | 60 | Fashion |
| Kitchen | 40 | Kitchen |
| Electronics | 130 | Electronics |
| Fashion | 50 | Fashion |
| Accessories | 100 | Accessories |

| Filter | |
|---|---|
| Backpack | Accesories |
| Sneakers | Fashion |
| Coffee Maker | Kitchen |
| Fitness Tracker | Electronics |

### *2. Correcting Inconsistent Data:*

- Inconsistent text formats present in the "Product Name" column. *(*Rectified by Find & Replace Method [Ctrl+H] )
- Typos present in the "Category" column.*(Rectified manually)*



| B | C | D | E | F |
|---|---|---|---|---|
| Product Nam | Brand Name | Price ($) | Quantity | Category |
| Laptop | Dell | 1000 | 30 | Electronics |
| Sneakers | Nike | 80 | 15 | Fashion |
| Coffee Maker | Keurig | 130 | 40 | Kitchen |
| Smartphone | Samsung | 900 | 25 | Electronics |
| Backpack | North Face | 70 | 20 | |
| Headphones | Sony | | 45 | Electronics |
| T-Shirt | Adidas | 30 | 5 | Fashion |
| Blender | Ninja | 90 | 35 | Kitchen |
| Tablet | Apple | 500 | 50 | Electronics |
| Hiking Boots | Timberland | 130 | 10 | Outdoor |
| Laptop | HP | 950 | 25 | Electronics |
| Sneakers | Adidas | 90 | 40 | |
| Coffee Maker | Nespresso | 120 | 35 | |
| Smartwatch | Fitbit | 150 | 15 | Electronics |
| Headphones | Bose | 250 | 20 | Electronics |
| Laptop Bag | Samsonite | 50 | 35 | Accessories |
| Smartwatch | Huawei | 160 | 15 | Electronics |
| Laptop | Asus | 980 | 10 | Electronics |
| Sunglasses | Oakley | 150 | 15 | Fashion |
| Camping Tent | Coleman | | 10 | Outdoor |
| Camera | Nikon | 700 | 50 | Electronics |
| Microwave | Panasonic | 80 | 20 | Kitchen |
| Fitness Tracker | Xiaomi | 150 | 30 | |
| Laptop Bag | Samsonite | 50 | 35 | Accessories |
| Smartphone | Google | 800 | 45 | Electronics |
| Sunglasses | Ray-Ban | | 25 | Fashion |
| Blender | Vitamix | 400 | 40 | Kitchen |
| Headphones | Bose | 250 | 20 | Electronics |
| Dress | Zara | 60 | 30 | Fashion |
| Toaster | Hamilton | 40 | 10 | Kitchen |
| Fitness Tracker | Garmin | 130 | 5 | Electronics |
| Jeans | Levi's | 50 | 50 | Fashion |
| Laptop | HP | 950 | 25 | Electronics |
| Watch | Casio | 100 | 20 | Accessories |

### *3. Removing Duplicates:*

- Rectified duplicate rows in the dataset.(Highlighting & Removing Duplicated Dataset in the “ Product ID”.)
- Highlighting duplicates dataset.

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | | | | | | |
| 2 | Product ID | Product Nam | Brand Name | Price ($) | Quantity | Category |
| 3 | 28-JAN-US | Laptop | Dell | 1000 | 30 | Electronics |
| 4 | 15-FEB-US | Sneakers | Nike | 80 | 15 | Fashion |
| 5 | 03-MAR-US | Coffee Maker | Keurig | 130 | 40 | Kitchen |
| 6 | 11-APR-US | Smartphone | Samsung | 900 | 25 | Electronics |
| 7 | 22-MAY-US | Backpack | North Face | 70 | 20 | |
| 8 | 07-JUN-UK | Headphones | Sony | | 45 | Electronics |
| 9 | 19-JUL-UK | T-Shirt | Adidas | 30 | 5 | Fashion |
| 10 | 23-AUG-UK | Blender | Ninja | 90 | 35 | Kitchen |
| 11 | 05-SEP-UK | Tablet | Apple | 500 | 50 | Electronics |
| 12 | 14-OCT-UK | Hiking Boots | Timberland | 130 | 10 | Outdoor |
| 13 | 17-JUN-IN | Laptop | HP | 950 | 25 | Electronics |
| 14 | 25-NOV-AU | Sneakers | Adidas | 90 | 40 | |
| 15 | 08-DEC-DE | Coffee Maker | Nespresso | 120 | 35 | |
| 16 | 18-FEB-CA | Smartwatch | Fitbit | 150 | 15 | Electronics |
| 17 | 16-APR-ES | Headphones | Bose | 250 | 20 | Electronics |
| 18 | 21-AUG-CA | Laptop Bag | Samsonite | 50 | 35 | Accessories |
| 19 | 20-AUG-CN | Smartwatch | Huawei | 160 | 15 | Electronics |
| 20 | 27-JAN-IT | Laptop | Asus | 980 | 10 | Electronics |
| 21 | 01-MAR-UK | Sunglasses | Oakley | 150 | 15 | Fashion |
| 22 | 14-AUG-US | Camping Tent | Coleman | | 10 | Outdoor |
| 23 | 14-MAY-RU | Camera | Nikon | 700 | 50 | Electronics |
| 24 | 09-JAN-CA | Microwave | Panasonic | 80 | 20 | Kitchen |
| 25 | 19-JUL-BR | Fitness Tracker | Xiaomi | 150 | 30 | |
| 26 | 21-AUG-CA | Laptop Bag | Samsonite | 50 | 35 | Accessories |
| 27 | 29-SEP-CA | Smartphone | Google | 800 | 45 | Electronics |
| 28 | 03-JUN-CA | Sunglasses | Ray-Ban | | 25 | Fashion |
| 29 | 11-JUL-CA | Blender | Vitamix | 400 | 40 | Kitchen |
| 30 | 16-APR-ES | Headphones | Bose | 250 | 20 | Electronics |
| 31 | 07-MAR-CA | Dress | Zara | 60 | 30 | Fashion |
| 32 | 13-APR-CA | Toaster | Hamilton | 40 | 10 | Kitchen |
| 33 | 24-MAY-CA | Fitness Tracker | Garmin | 130 | 5 | Electronics |
| 34 | 02-DEC-CA | Jeans | Levi's | 50 | 50 | Fashion |
| 35 | 17-JUN-IN | Laptop | HP | 950 | 25 | Electronics |
| 36 | 09-JUL-FR | Watch | Casio | 100 | 20 | Accessories |
| 37 | | | | | | |

- Removed Duplicates Values.

| Product ID | Product Name | Brand Name | Price ($) | Quantity | Category |
|---|---|---|---|---|---|
| 28-JAN-US | Laptop | Dell | 1000 | 30 | Electronics |
| 15-FEB-US | Sneakers | Nike | 80 | 15 | Fashion |
| 03-MAR-US | Coffee Maker | Keurig | 130 | 40 | Kitchen |
| 11-APR-US | Smartphone | Samsung | 900 | 25 | Electronics |
| 22-MAY-US | Backpack | North Face | 70 | 20 | |
| 07-JUN-UK | Headphones | Sony | | 45 | Electronics |
| 19-JUL-UK | T-Shirt | Adidas | 30 | 5 | Fashion |
| 23-AUG-UK | Blender | Ninja | 90 | 35 | Kitchen |
| 05-SEP-UK | Tablet | Apple | 500 | 50 | Electronics |
| 14-OCT-UK | Hiking Boots | Timberland | 130 | 10 | Outdoor |
| 17-JUN-IN | Laptop | HP | 950 | 25 | Electronics |
| 25-NOV-AU | Sneakers | Adidas | 90 | 40 | |
| 08-DEC-DE | Coffee Maker | Nespresso | 120 | 35 | |
| 18-FEB-CA | Smartwatch | Fitbit | 150 | 15 | Electronics |
| 16-APR-ES | Headphones | Bose | 250 | 20 | Electronics |
| 21-AUG-CA | Laptop Bag | Samsonite | 50 | 35 | Accessories |
| 20-AUG-CN | Smartwatch | Huawei | 160 | 15 | Electronics |
| 27-JAN-IT | Laptop | Asus | 980 | 10 | Electronics |
| 01-MAR-UK | Sunglasses | Oakley | 150 | 15 | Fashion |
| 14-AUG-US | Camping Tent | Coleman | | 10 | Outdoor |
| 14-MAY-RU | Camera | Nikon | 700 | 50 | Electronics |
| 09-JAN-CA | Microwave | Panasonic | 80 | 20 | Kitchen |
| 19-JUL-BR | Fitness Tracker | Xiaomi | 150 | 30 | |
| 29-SEP-CA | Smartphone | Google | 800 | 45 | Electronics |
| 03-JUN-CA | Sunglasses | Ray-Ban | | 25 | Fashion |
| 11-JUL-CA | Blender | Vitamix | 400 | 40 | Kitchen |
| 07-MAR-CA | Dress | Zara | 60 | 30 | Fashion |
| 13-APR-CA | Toaster | Hamilton | 40 | 10 | Kitchen |
| 24-MAY-CA | Fitness Tracker | Garmin | 130 | 5 | Electronics |
| 02-DEC-CA | Jeans | Levi's | 50 | 50 | Fashion |
| 09-JUL-FR | Watch | Casio | 100 | 20 | Accessories |

***4. Splitting and Merging Data:***

Used Flash Fill to separate the Manufacturing date,Country Code and merge Brand name and Product name into the Separated Column.

| | I | J | K |
|---|---|---|---|
| 1 | USED (Flash Fill) | | |
| 2 | Manufacturing date | Country Code | Product Brand |
| 3 | 28-JAN | US | Dell Laptop |
| 4 | 12-FEB | US | Nike Sneakers |
| 5 | 03-MAR | US | Keurig Coffee Maker |
| 6 | 11-APR | US | Samsung Smartphone |
| 7 | 22-MAY | US | North Face Backpack |
| 8 | 07-JUN | UK | Sony Headphones |
| 9 | 19-JUL | UK | Adidas T-Shirt |
| 10 | 23-AUG | UK | Ninja Blender |
| 11 | 05-SEP | UK | Apple Tablet |
| 12 | 14-OCT | UK | Timberland Hiking Boots |
| 13 | 17-JUN | IN | HP Laptop |
| 14 | 25-NOV | AU | Adidas Sneakers |
| 15 | 08-DEC | DE | Nespresso Coffee Maker |
| 16 | 18-FEB | CA | Fitbit Smartwatch |
| 17 | 16-APR | ES | Bose Headphones |
| 18 | 21-AUG | CA | Samsonite Laptop Bag |
| 19 | 20-AUG | CN | Huawei Smartwatch |
| 20 | 27-JAN | IT | Asus Laptop |
| 21 | 01-MAR | UK | Oakley Sunglasses |
| 22 | 14-AUG | US | Coleman Camping Tent |
| 23 | 14-MAY | RU | Nikon Camera |
| 24 | 09-JAN | CA | Panasonic Microwave |
| 25 | 19-JUL | BR | Xiaomi Fitness Tracker |
| 26 | 29-SEP | CA | Google Smartphone |
| 27 | 03-JUN | CA | Ray-Ban Sunglasses |
| 28 | 11-JUL | CA | Vitamix Blender |
| 29 | 07-MAR | CA | Zara Dress |
| 30 | 13-APR | CA | Hamilton Toaster |
| 31 | 24-MAY | CA | Garmin Fitness Tracker |
| 32 | 02-DEC | CA | Levi's Jeans |
| 33 | 09-JUL | FR | Casio Watch |
| 34 | | | |

***5. Number Formatting:***

- The data type formatted price currency instead of price.
- The Manufacturing data formatted as a Date Format.

| | | USED (F |
|---|---|---|
| Price Missing Value | Category Missing Value | Manufacturing date |
| £1,000.00 | Electronics | 28 |
| £80.00 | Fashion | 15 |
| £130.00 | Kitchen | 03 |
| £900.00 | Electronics | 11 |
| £70.00 | Accesories | 22 |
| £250.00 | Electronics | 07 |
| £30.00 | Fashion | 19 |
| £90.00 | Kitchen | 23 |
| £500.00 | Electronics | 05 |
| £130.00 | Outdoor | 14 |
| £950.00 | Electronics | 17 |
| £90.00 | Fashion | 25 |
| £120.00 | Kitchen | 08 |
| £150.00 | Electronics | 18 |
| £250.00 | Electronics | 16 |
| £50.00 | Accessories | 21 |
| £160.00 | Electronics | 20 |
| £980.00 | Electronics | 27 |
| £150.00 | Fashion | 01 |
| No Price Available | Outdoor | 14 |
| £700.00 | Electronics | 14 |
| £80.00 | Kitchen | 09 |
| £150.00 | Electronics | 19 |
| £800.00 | Electronics | 29 |
| £150.00 | Fashion | 03 |
| £400.00 | Kitchen | 11 |
| £60.00 | Fashion | 07 |
| £40.00 | Kitchen | 13 |
| £130.00 | Electronics | 24 |
| £50.00 | Fashion | 02 |
| £100.00 | Accessories | 09 |




| J | K | L | M |
|---|---|---|---|
| Flash Fill) | | | |
| Column1 | Manufacturing Date 2 | Country Code | Product Brand |
| JAN | 28-Jan-2026 | US | Dell Laptop |
| FEB | 15-Feb-2026 | US | Nike Sneakers |
| MAR | 03-Mar-2026 | US | Keurig Coffee Maker |
| APR | 11-Apr-2026 | US | Samsung Smartphone |
| MAY | 22-May-2026 | US | North Face Backpack |
| JUN | 07-Jun-2026 | UK | Sony Headphones |
| JUL | 19-Jul-2026 | UK | Adidas T-Shirt |
| AUG | 23-Aug-2026 | UK | Ninja Blender |
| SEP | 05-Sep-2026 | UK | Apple Tablet |
| OCT | 14-Oct-2026 | UK | Timberland Hiking Boots |
| JUN | 17-Jun-2026 | IN | HP Laptop |
| NOV | 25-Nov-2026 | AU | Adidas Sneakers |
| DEC | 08-Dec-2026 | DE | Nespresso Coffee Maker |
| FEB | 18-Feb-2026 | CA | Fitbit Smartwatch |
| APR | 16-Apr-2026 | ES | Bose Headphones |
| AUG | 21-Aug-2026 | CA | Samsonite Laptop Bag |
| AUG | 20-Aug-2026 | CN | Huawei Smartwatch |
| JAN | 27-Jan-2026 | IT | Asus Laptop |
| MAR | 01-Mar-2026 | UK | Oakley Sunglasses |
| AUG | 14-Aug-2026 | US | Coleman Camping Tent |
| MAY | 14-May-2026 | RU | Nikon Camera |
| JAN | 09-Jan-2026 | CA | Panasonic Microwave |
| JUL | 19-Jul-2026 | BR | Xiaomi Fitness Tracker |
| SEP | 29-Sep-2026 | CA | Google Smartphone |
| JUN | 03-Jun-2026 | CA | Ray-Ban Sunglasses |
| JUL | 11-Jul-2026 | CA | Vitamix Blender |
| MAR | 07-Mar-2026 | CA | Zara Dress |
| APR | 13-Apr-2026 | CA | Hamilton Toaster |
| MAY | 24-May-2026 | CA | Garmin Fitness Tracker |
| DEC | 02-Dec-2026 | CA | Levi's Jeans |
| JUL | 09-Jul-2026 | FR | Casio Watch |

### *6. Conditional Formatting:*

- I used data bars for the price column.
- I highlighted the cells containing “Electronics”.

| | C | D | E | F | G | H |
|---|---|---|---|---|---|---|
| 1 | | | | | | |
| 2 | | Price ($) | Quantity | Category | Price Missing Value | Category Missing Value |
| 3 | | £1,000.00 | 30 | Electronics | 1000 | Electronics |
| 4 | | £80.00 | 15 | Fashion | 80 | Fashion |
| 5 | | £130.00 | 40 | Kitchen | 130 | Kitchen |
| 6 | | £900.00 | 25 | Electronics | 900 | Electronics |
| 7 | | £70.00 | 20 | | 70 | Accesories |
| 8 | | | 45 | Electronics | 250 | Electronics |
| 9 | | £30.00 | 5 | Fashion | 30 | Fashion |
| 10 | | £90.00 | 35 | Kitchen | 90 | Kitchen |
| 11 | | £500.00 | 50 | Electronics | 500 | Electronics |
| 12 | | £130.00 | 10 | Outdoor | 130 | Outdoor |
| 13 | | £950.00 | 25 | Electronics | 950 | Electronics |
| 14 | | £90.00 | 40 | | 90 | Fashion |
| 15 | | £120.00 | 35 | | 120 | Kitchen |
| 16 | | £150.00 | 15 | Electronics | 150 | Electronics |
| 17 | | £250.00 | 20 | Electronics | 250 | Electronics |
| 18 | | £50.00 | 35 | Accessories | 50 | Accessories |
| 19 | | £160.00 | 15 | Electronics | 160 | Electronics |
| 20 | | £980.00 | 10 | Electronics | 980 | Electronics |
| 21 | | £150.00 | 15 | Fashion | 150 | Fashion |
| 22 | | | 10 | Outdoor | No Price Available | Outdoor |
| 23 | | £700.00 | 50 | Electronics | 700 | Electronics |
| 24 | | £80.00 | 20 | Kitchen | 80 | Kitchen |
| 25 | | £150.00 | 30 | | 150 | Electronics |
| 26 | | £800.00 | 45 | Electronics | 800 | Electronics |
| 27 | | | 25 | Fashion | 150 | Fashion |
| 28 | | £400.00 | 40 | Kitchen | 400 | Kitchen |
| 29 | | £60.00 | 30 | Fashion | 60 | Fashion |
| 30 | | £40.00 | 10 | Kitchen | 40 | Kitchen |
| 31 | | £130.00 | 5 | Electronics | 130 | Electronics |
| 32 | | £50.00 | 50 | Fashion | 50 | Fashion |
| 33 | | £100.00 | 20 | Accessories | 100 | Accessories |
| 34 | | | | | | |